

Hybrid Fourier Neural Operator–Lattice Boltzmann Method[☆]

A. Junk^{*}, J.M. Winter, M. Tütken¹, S. Schmidt, N.A. Adams

Technical University of Munich, Department Engineering Physics and Computation, Chair of Aerodynamics and Fluid Mechanics, Boltzmannstraße 15, Garching, 85748, Bavaria, Germany

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ABSTRACT

We propose an accelerated computational fluid dynamics framework based on a hybrid Fourier Neural Operator–Lattice Boltzmann Method (FNO–LBM) for steady and unsteady weakly compressible flows. The approach is evaluated on two-dimensional, single-phase benchmark problems at low Reynolds numbers using an SRT LBM formulation. For steady porous media flows, FNO-based initialization significantly accelerates convergence across macroscopic fields, achieving up to 87% speed-up for density and 55% for pressure-drop depending on the convergence tolerance, while preserving steady-state accuracy. Simulations of unsteady flows can be accelerated by hybrid coupling strategies that employ FNO rollouts embedded into LBM time advancement in a way of super-time-stepping, where one FNO step substitutes several LBM steps at once. Global and time-resolved error metrics across 100 trajectories for generic 2D flows demonstrate that hybridization consistently improves accuracy and stabilizes long-horizon rollouts relative to purely autoregressive surrogates. Best efficiency is achieved for a lightweight 2.6M-parameter FNO, which diverges under pure autoregressive rollout but achieves up to 99% error reduction under hybrid coupling, matching the predictive capability of a much more expensive 11.2M-parameter model. The hybrid framework enhances predictive fidelity, suppresses error accumulation, and enables small and cheap surrogate models to operate effectively within the same error regime as larger surrogates. Overall, the results demonstrate that hybrid FNO–LBM coupling can improve stability and efficiency in controlled low-Re 2D settings, enabling compact surrogate models to operate effectively within the accuracy range of larger models and the underlying LBM solver.

1. Introduction

High-fidelity numerical simulation of fluid flows remains central to fluid dynamics analysis, optimization, and design across science and engineering. However, the computational cost associated with resolving complex geometries, multiscale features, and long-time unsteady dynamics continues to pose significant challenges. The Lattice Boltzmann Method (LBM) offers a robust, efficient and accurate computational framework for weakly compressible flows, but as with all classical CFD methods its cost scales with grid resolution and simulation time, limiting applications to problems that require rapid turnaround, extensive parameter studies, or real-time feedback [1]. These constraints have motivated growing interest in machine learning surrogate approaches capable of accelerating CFD simulations within specific regimes while preserving physical accuracy [2].

With data-driven machine-learning (ML) models for flow evolution, surrogate modeling has therefore emerged as a promising pathway for accelerating fluid simulations [3–7]. However, many conventional neural network architectures operate on fixed spatial grids and struggle to generalize across discretizations or maintain stability during long autoregressive rollouts, restricting their reliability in high-resolution or long-horizon unsteady simulations. Neural operators address these limitations by learning mappings between function spaces rather than discrete arrays, enabling resolution-invariant inference and single-pass prediction of entire flow fields [8–11]. Among them, the Fourier Neural Operator (FNO) has emerged as a prominent architecture due to its spectral convolution formulation and quasi-linear scaling with resolution [7]. FNOs and their variants – such as U-FNO [12], F-FNO [13], SFNO [14], and multi-scale extensions [15] – have demonstrated impressive performance across a wide range of physics-informed

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^{*} Corresponding author.

E-mail addresses: alexandra.junk@tum.de (A. Junk), josef.winter@tum.de (J.M. Winter), meike.tuetken@siemens.com (M. Tütken), steffen.schmidt@tum.de (S. Schmidt), nikolaus.adams@tum.de (N.A. Adams).

¹ Present address: Siemens Industry Software NV, Interleuvenlaan 68, 3001 Heverlee, Belgium; KU Leuven, Department of Mechanical Engineering, Celestijnenlaan 300, 3001 Heverlee, Belgium.

tasks, including porous media flows [16], elastodynamics [17], hydraulic tomography [18], nonlinear optics [19], and tokamak plasma evolution [20], establishing operator learning as a versatile tool for approximation of high-dimensional PDE solutions.

Despite their expressiveness, computationally cheap FNO models may rapidly accumulate error during autoregressive rollout and diverge after only a few autoregressive steps [13,14,20]. This limitation has prompted increasing interest in hybrid modeling frameworks that combine machine learning surrogates with numerical solvers. By intermittently invoking directly a consistent approximation of the governing equations, hybrid approaches can correct accumulated errors, enforce physical constraints, and stabilize long-term predictions. Several such hybrid CFD frameworks have recently emerged. In the context of LBM, Rabbani and Babaei [21] proposed a machine-learning-accelerated permeability workflow, Zhang et al. [22] combined ML and LBM for heat-flux prediction in phase-change processes, and Fischer et al. [23] used a reinforcement-learning closure model to stabilize under-resolved LBM simulations. The latter is particularly related to the present work in that machine learning is used to enhance the stability of LBM rollouts; however, in contrast to Fischer et al. [23], who learn closures within the LBM evolution, we instead use LBM as a corrective mechanism to stabilize autoregressive ML predictions.

Recent work has focused more directly on learning kinetic or mesoscopic representations of the LBM dynamics. Corbetta et al. [24] introduced Lattice-Equivariant Neural Networks (LENNs) to approximate the collision operator while explicitly enforcing the discrete symmetries of the lattice. Their results highlight the importance of symmetry preservation for stability and accuracy in learned kinetic operators. Similarly, Xue et al. [25] proposed a physics-informed neural operator framework that predicts the full set of particle distribution functions and learns the forward kinetic evolution over extended time horizons, incorporating moment-matching constraints and global equivariance to maintain physical consistency across scales. Both approaches operate at the level of particle distribution functions and therefore remain tightly coupled to the underlying lattice structure and kinetic formulation of the method, either through explicit collision modeling or through distribution-level operator learning.

In contrast, we deliberately adopt a macroscopic-field surrogate that predicts hydrodynamic quantities rather than particle populations. This design choice decouples the learned model from the specific collision operator, lattice discretization, and even from the LBM formulation itself, enabling a more flexible hybrid coupling. In particular, our approach allows the surrogate to be trained on macroscopic data from different numerical solvers or LBM collision and lattice configurations and reused across discretizations, while relying on LBM only as a corrective mechanism to enforce consistency and stabilize long-term rollouts. This distinction places our method within a complementary class of hybrid models that prioritize modularity and solver-agnostic coupling over tight integration with kinetic-level dynamics.

More broadly, hybrid ML-numerical solver systems include FNO-coupled finite-difference solvers for fluid-structure interaction [26], neural networks used as pressure solvers or closure models in Navier-Stokes flows [27] and DeepONet preconditioning [28]. These developments suggest that hybridization can combine the speed of ML surrogates with the robustness of physics-based solvers, but the degree to which hybrid frameworks can compensate for limited surrogate capacity or enforce long-horizon stability remains insufficiently explored [2].

In this work, we address this gap by investigating hybrid integration of FNOs within an LBM framework for both steady and unsteady weakly compressible flows, specifically focusing on two-dimensional, single-phase flows at low Reynolds numbers, which provide a controlled setting for assessing hybrid surrogate-solver coupling. First, we employ FNO-predicted macroscopic fields for acceleration of steady-state LBM simulations of porous media flows. Second, we consider a two-dimensional double shear layer configuration at low Reynolds number,

which is highly sensitive to accumulated errors and thus is a critical test of temporal surrogate stability and for assessing hybrid strategies in autoregressive rollout. In this case, the FNO is employed as a super-time-stepping alike algorithm that replaces a large number of LBM time steps with one FNO rollout-step. We compare a large 11.2M-parameter FNO with a lightweight 2.6M-parameter model and find that numerical corrections are crucial for stabilization of the lightweight model during autoregressive rollout. Our contributions are threefold:

- **Steady-state-convergence acceleration:** FNO-initialized LBM achieves substantial reductions in convergence time, reducing iteration counts by up to 87% for density and 55% for pressure-drop metrics depending on the convergence tolerance, with no loss of accuracy relative to converged steady-states obtained with LBM only.
- **Improvement of autoregressive FNO rollouts by hybridization:** Hybrid FNO-LBM coupling significantly improves accuracy even for autoregressively stable FNO rollouts by up to 60% and decreases total simulation runtimes by up to 11.8%.
- **Stabilization of low-cost FNO surrogates by hybridization:** Hybridization enables a 2.6M-parameter FNO, which is unstable as a standalone surrogate, to achieve 99% error reduction and approach the accuracy of a significantly larger 11.2M-parameter surrogate.

These results demonstrate that hybrid neural-operator frameworks offer a powerful and computationally efficient means of accelerating LBM in the considered low-Reynolds-number 2D setting while maintaining physical fidelity and transforming compact surrogates into physically reliable models for long-horizon unsteady prediction.

2. Methodology

2.1. The lattice Boltzmann method

All simulations are run in single-precision using TorchLBM [29], a fully-differentiable and modular PyTorch implementation of an LBM solver for the weakly compressible Navier-Stokes Equations (NSE). LBM is a mesoscopic numerical method [30] that computes the evolution of discrete particle distribution functions f_i along predefined discrete lattice velocities $\vec{c}_i$ over a structured lattice grid.

The general simulation loop of TorchLBM is depicted in Fig. 1 and consists of six recurrent modules and one initialization block at the start of the simulation. After initialization of the macroscopic density ρ and velocity $\vec{u}$, the main simulation loop starts with the calculation of the equilibrium distribution f_i^{eq} following

$$f_i^{eq} = w_i \rho \left(1 + \frac{\vec{u} \cdot \vec{c}_i}{c_s^2} + \frac{(\vec{u} \cdot \vec{c}_i)^2}{2c_s^4} - \frac{\vec{u} \cdot \vec{u}}{2c_s^2} \right), \quad (1)$$

where w_i denotes the respective lattice weights and c_s^2 the isothermal model's speed of sound. The number and configuration of lattice velocities and weights follow the $DdQq$ convention, with d spatial dimensions and q discrete velocities. For 2D configurations best efficiency is reached with the $D2Q9$ scheme.

The equilibrium calculation is followed by the two split operations of the LBM algorithm: collision and streaming. During the collision step, local non-linear particle interactions are modeled by the collision operator. For simplicity, we adopt here the Single Relaxation Time (SRT) collision model

$$f_i^*(\vec{x}, t) = f_i(\vec{x}, t) - \frac{\Delta t}{\tau} (f_i - f_i^{eq}(\vec{x}, t)) \quad (2)$$

with the relaxation time τ . In the streaming step, post-collision distributions f_i^* are propagated through a linear, non-local streaming operation to neighboring nodes following

$$f_i(\vec{x} + \vec{c}_i \Delta t, t + \Delta t) = f_i^*(\vec{x}, t). \quad (3)$$

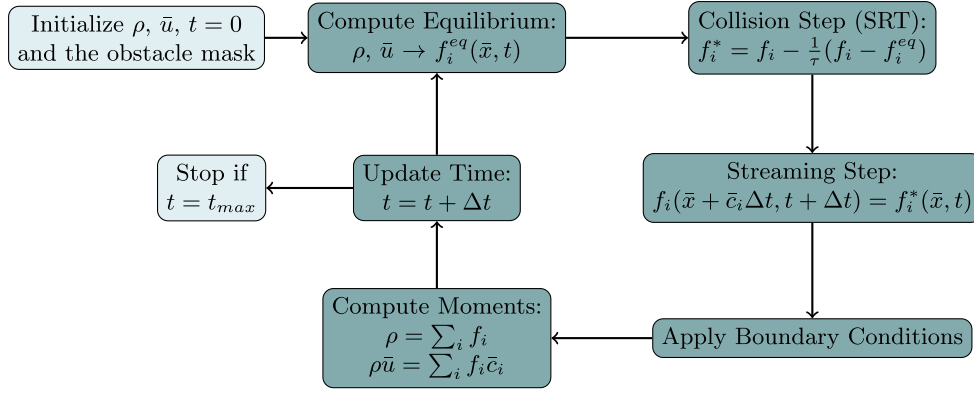


Fig. 1. Visualization of the LBM simulation loop. If no macroscopic fields are available, they are set to default $\rho = 1.0 \frac{\text{kg}}{\text{m}^3}$, $\bar{u} = 0.0 \frac{\text{m}}{\text{s}}$ and no obstacle mask.

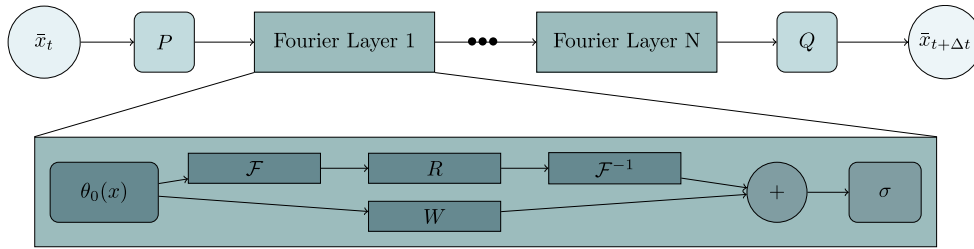


Fig. 2. FNO Architecture with the lifting network P , the Fourier transform $\mathcal{F}$, the learnable filter R , the local linear transform W , the activation σ and the projection Q back to physical space.

Lastly, the macroscopic quantities are recovered from distribution moments

$$\rho(\bar{x}, t) = \sum_i f_i(\bar{x}, t), \quad (4)$$

$$\rho \bar{u}(\bar{x}, t) = \sum_i \bar{c}_i f_i(\bar{x}, t), \quad (5)$$

and the simulation time is updated.

A comprehensive discussion of the LBM and its numerical implementation is provided e.g. by Krüger et al. [1], Succi [31] and Lallemand et al. [32].

The macroscopic density and velocity fields serve as the input data for training and validation of the FNO models. Although LBM provides access to both mesoscopic and macroscopic states, the exclusive use of macroscopic quantities improves the generality of the proposed hybrid framework. Specifically, it allows integration with surrogate models trained on macroscopic data obtained from other solvers and supports seamless transfer across different LBM setups, independent of the collision operator and lattice discretization employed.

2.2. Fourier Neural Operator

Li et al. [33] introduced the FNO as a discretization-invariant architecture that learns mappings between infinite-dimensional function spaces. The architecture builds on the Neural Operator framework proposed by Kovachki et al. [34], replacing the kernel integral operator with a global convolution in Fourier space, leading to the computationally efficient approximation. The architectural structure of the FNO used in this study is illustrated in Fig. 2.

Let $\bar{x}_t \in \mathbb{R}^d$ denote the input function, in our case the density and velocity fields $\bar{x}_t = (u_t, v_t, \rho_t)$ at time t . This input function is first lifted into latent space by the shallow fully-connected neural network P via

$P(\bar{x}_t) = \theta_0(x)$. The FNO then performs iterative updates $\theta_n \mapsto \theta_{n+1}$ for $n = 0, 1, \dots, N-1$ across the N Fourier layers following

$$\theta_{n+1}(x) = \sigma \left(W \theta_n(x) + \mathcal{F}^{-1} \left(R(\bar{k}) \cdot \mathcal{F}(\theta_n)(\bar{k}) \right) \right), \quad (6)$$

where σ is a pointwise nonlinearity (GELU), W is a local linear transformation, $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the Fourier and inverse Fourier transforms, $\bar{k}$ represents the cut-off filtered frequency modes, and R is a learnable complex-valued filter kernel applied to each Fourier mode.

Lastly, the output is projected back to physical space with $Q(\theta_{N-1}) = \bar{x}_{t+\Delta t}$, where in our case $\Delta t = \Delta T_{FNO}$ represents the discrete time-step size of the FNO.

2.3. Hybrid FNO-LBM framework

We propose a hybrid framework that couples a fully trained FNO with the LBM solver, thereby exploiting the complementary strengths of both approaches. Within this framework, the FNO is employed either as a generator of an initial flow field for steady-state problems or during time marching, functioning as a super-time-stepping surrogate for macroscopic fields alternatively with direct LBM time steps. This procedure leverages the computational efficiency of the ML model while maintaining physical fidelity through the recurrent physical bias of the discretized evolution equations. Although the present work focuses on FNO-LBM coupling, the proposed framework is not restricted to operator-learning models and can, in principle, be combined with other surrogate architectures. To demonstrate this flexibility, D investigates the use of a residual U-Net as an alternative surrogate for time integration.

For steady-state problems, the FNO serves an initial guess for the macroscopic fields, which is passed to the LBM solver. The solver then proceeds to converge to the steady solution by LBM time marching, without further interaction with the ML model (see Fig. 3(a)).

For unsteady flows, the framework alternates between FNO-based predictions and LBM time stepping throughout the entire simulation horizon (see Fig. 3(b)). The switching frequency between FNO and LBM is governed by a predefined temporal criterion. All FNOs employed in this work are trained to predict a fixed time increment of

$$\Delta T_{FNO} = \alpha \cdot \Delta t_{LBM} = \Delta T, \quad (7)$$

where Δt_{LBM} represents the simulation time step and α is a configurable stride parameter that can be adjusted depending on the problem setup and desired trade-off between accuracy and speed-up. For the double shear layer configuration considered in this study, where $L = 1$ m and $U_\infty = 1$ m/s lead to the convective time scale $L/U_\infty = 1$ s, we choose $\Delta T_{FNO} = 0.1$ s, which yields $\alpha = 223$ following

$$\Delta T_{FNO} = 223 \Delta t_{LBM} = 0.1 \text{ s} = 0.1(L/U_\infty) \quad (8)$$

i.e., 10% of a convective time unit. This provides a direct physical interpretation of the surrogate as advancing the flow over a finite, yet subdominant fraction of the characteristic flow-through time. The choice $\Delta T_{FNO} = 223 \cdot \Delta t_{LBM}$ arises from the temporal spacing of training data snapshots and reflects a practical trade-off between computational acceleration and predictive accuracy, where smaller strides reduce the achievable speed-up in the hybrid setting and larger strides can become harder to learn accurately. It should be emphasized that the present configuration is relatively conservative from the perspective of computational acceleration, as each surrogate prediction bypasses only 223 LBM time steps. For simulations requiring finer temporal resolutions (smaller Δt_{LBM}), the same physical prediction interval ΔT_{FNO} would correspond to substantially larger values of α , allowing a single FNO evaluation to replace a greater number of LBM updates and thereby yielding larger overall speed-ups. While the underlying LBM solver is subject to CFL and low-Mach stability constraints, the FNO operates on learned state transitions and is therefore not directly constrained by these conditions. However, increasing ΔT_{FNO} effectively enlarges the temporal extrapolation horizon, which impacts accuracy even in the absence of explicit numerical stability limits. In this sense, the chosen value represents a robust operating point rather than a strict upper bound.

The LBM advances the solution over multiple of its intrinsic time steps following

$$\Delta T_{LBM} = k \cdot \Delta T_{FNO} = k \cdot \Delta T, \quad k \in \{1, 2, 3\}. \quad (9)$$

Based on this relation, we define the following hybrid configurations:

- 1:1-Hybrid: ΔT predicted by the FNO and ΔT advanced by the LBM ($k = 1$),
- 1:2-Hybrid: ΔT predicted by the FNO and $2\Delta T$ advanced by the LBM ($k = 2$),
- 1:3-Hybrid: ΔT predicted by the FNO and $3\Delta T$ advanced by the LBM ($k = 3$).

These configurations enable systematic control over the degree of ML intervention and thereby provide a tunable balance between computational speed-up and numerical accuracy.

The use of a fixed temporal switching criterion constitutes a deliberately simple hybridization strategy. In contrast, much of the recent literature on ML-solver coupling focuses on embedding learned models directly into the numerical method and training them jointly with the solver dynamics. For example, Kochkov et al. [35] embed learned interpolation and correction operators within the numerical discretization itself and optimize them end-to-end as part of the CFD solver. This approach improves the accuracy of coarse-grid simulations while maintaining stable long-term rollouts through solver-consistent training.

Similarly, Um et al. [36] introduce a solver-in-the-loop framework in which neural networks are trained through recurrent interaction

with a differentiable PDE solver. By exposing the model to the consequences of its own predictions during training, the learned correction remains consistent with the evolving numerical solution, thereby improving stability and reducing long-term error accumulation. Building on this paradigm, List et al. [37] integrate learned turbulence models into a differentiable flow solver and demonstrate that training over increasingly long unrolled solver trajectories leads to more accurate and stable hybrid simulations. Their results further underline the importance of accounting for temporal error propagation during training and of incorporating physically motivated objectives into the optimization process.

In contrast to these tightly coupled approaches, the present work adopts a modular runtime-coupling strategy. The surrogate model is trained independently of the numerical solver and interacts with the LBM exclusively during inference through periodic alternation between surrogate prediction and numerical time integration. Consequently, the LBM is not employed to improve surrogate training, but instead acts as a recurrent physical correction mechanism that mitigates autoregressive error accumulation. This design avoids the need for differentiable solvers and solver-specific training procedures while preserving consistency with the underlying LBM dynamics through periodic numerical correction.

Accordingly, the objective of the present study differs from that of existing solver-in-the-loop approaches. Rather than learning solver-aware discretizations, correction operators, or turbulence closures, we investigate the stabilization of a pre-trained surrogate through intermittent interaction with a physics-based solver. In particular, the correction frequency itself becomes a controllable design parameter. The proposed hybrid configurations therefore enable a systematic investigation of the trade-off between surrogate-driven acceleration and solver-based stabilization. From this perspective, the framework provides a complementary route toward hybrid ML-CFD methods that emphasizes modularity, portability, and solver-agnostic deployment.

While adaptive switching strategies based on local residuals, conservation metrics, or spectral diagnostics could further improve efficiency by triggering solver corrections only when necessary, the design of robust and computationally inexpensive indicators remains an open challenge. The development of such adaptive correction criteria therefore constitutes an important direction for future work.

2.4. Field-to-distribution reconstruction

After each FNO prediction, the macroscopic fields $(\rho, \mathbf{u})$ are used to reconstruct the particle distribution functions required by the LBM solver. In this work, a standard equilibrium reconstruction $f_i = f_i^{eq}$ following Eq. (1) is deployed. This reconstruction removes the non-equilibrium contribution $f_i^{neq} = f_i - f_i^{eq}$, which contains the viscous stress information of the LBM formulation. Following reconstruction, the non-equilibrium moments are gradually re-established through the subsequent streaming and collision operations. The magnitude and influence of the removed non-equilibrium contribution depend on the relaxation time τ .

For $\tau < 1$, corresponding to low-viscosity flows, the SRT collision operator relaxes the distribution functions strongly toward equilibrium. After reconstruction, streaming rapidly generates new non-equilibrium moments and the influence of the reconstruction is expected to decay over a relatively small number of LBM time steps.

For $\tau = 1$, the post-collision state of the SRT model coincides exactly with the equilibrium distribution. In this case, replacing the distribution functions by their equilibrium values is equivalent to performing a complete relaxation step. The removed non-equilibrium stresses are subsequently rebuilt through the normal evolution of the flow.

For $\tau > 1$, corresponding to higher viscosities, the collision operator relaxes the distributions more gradually, as the magnitude of the physically consistent non-equilibrium contribution increases with viscosity

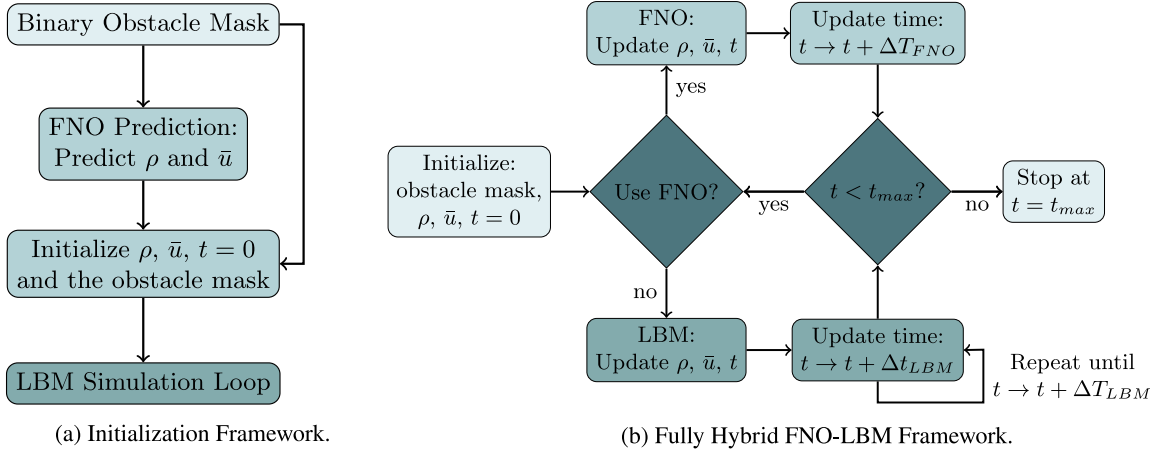


Fig. 3. Hybrid FNO-LBM Framework. For steady-state flows, the FNO provides an initial guess followed by the pure LBM Simulation loop depicted in Fig. 1. For non-steady flows, the switch criterion between LBM and FNO is defined as a fixed temporal switch based on a predefined time step.

and therefore with increasing relaxation time. Resetting the distributions to equilibrium therefore removes a larger amount of viscous stress information. As a consequence, the influence of the reconstruction may persist over a larger number of LBM time steps before the physically consistent non-equilibrium contribution is re-established.

The two benchmark configurations considered in this work employ different relaxation times and different modes of interaction between the FNO and the LBM solver. For the porous-media configuration ($\tau = 0.72$), the FNO is used exclusively to generate an initial condition. The equilibrium reconstruction is therefore performed only once at the beginning of the simulation. The subsequent LBM evolution naturally re-establishes the corresponding non-equilibrium moments, and any reconstruction-induced perturbation diminishes as the flow converges toward its steady state.

In contrast, the shear-layer configuration ($\tau = 0.52$) repeatedly alternates between FNO prediction and LBM integration. Each switching event requires reconstruction of the particle distribution functions and therefore resets the non-equilibrium contribution to zero. As a result, the physically consistent non-equilibrium stresses must be re-established after every FNO prediction through the subsequent streaming-collision dynamics. Owing to the low relaxation time, deviations from equilibrium are strongly damped by the BGK collision operator, while spatial gradients rapidly generate new non-equilibrium contributions. The shear-layer case therefore constitutes a stringent test of the hybrid coupling strategy, since equilibrium reconstruction is applied repeatedly throughout the simulation in a low-viscosity regime.

To isolate the error introduced by a single FNO-to-LBM transfer, the reconstructed particle populations f^{rec} are compared to the corresponding reference LBM populations f^{LBM} immediately after a reconstruction before any subsequent streaming or collision operations are performed. Reconstruction accuracy is assessed using the mean-squared error (MSE), the relative $\mathcal{L}_2$ error,

$$\mathcal{L}_2 = \frac{\|f^{rec} - f^{LBM}\|_2}{\|f^{LBM}\|_2}, \quad (10)$$

and the maximum absolute population error.

To quantify the accuracy of the equilibrium reconstruction, we evaluate the mean and standard deviation of these metrics over all switching events within a representative trajectory of the shear-layer benchmark. The reconstruction yields an MSE of $1.95 \times 10^{-8} \pm 1.77 \times 10^{-8}$, a relative $\mathcal{L}_2$ error of $8 \times 10^{-4} \pm 4 \times 10^{-4}$, and a maximum absolute error of $8 \times 10^{-4} \pm 3 \times 10^{-4}$.

These results indicate that the reconstructed populations differ from the reference distributions by less than 0.1% in terms of the relative $\mathcal{L}_2$ norm. Consequently, the equilibrium reconstruction introduces only a very small perturbation of the mesoscopic state at switching points. The

observed accuracy further suggests that the dominant source of error in the hybrid framework is not the field-to-distribution reconstruction itself, but rather the subsequent accumulation of surrogate prediction errors and their interaction with the LBM collision-streaming dynamics during repeated hybrid rollouts.

3. Results

This section presents results for (i) steady-state LBM initialization using FNO surrogates and (ii) unsteady hybrid FNO-LBM simulations. Training, simulations and rollouts are performed on an NVIDIA RTX A4000. FNO models are trained using the open-source NVIDIA PhysXNeMo framework [38], with default hyperparameters unless otherwise specified. Separate networks are trained for density and velocity prediction. Evaluation trajectories are generated using independent random seeds but within the same parameter ranges as the training data. Consequently, the test cases represent unseen realizations of the same underlying distribution and assess generalization across stochastic variability rather than across changes in the physical parameter space. At the same time, both benchmark configurations include evaluation on spatial resolutions that are not observed during training. Unless stated otherwise, the FNO models are trained exclusively on data generated at the baseline resolution and are subsequently evaluated on finer grids during inference. The reported results therefore include a resolution-transfer component in addition to conventional test-set generalization. A systematic assessment under stronger distribution shifts, such as variations in porosity, Reynolds number, obstacle statistics, or flow parameters outside the training range, is beyond the scope of the present work and is left for future investigation. Metrics are computed per field and per time step when applicable, global metrics are obtained via temporal averaging. Arithmetic means and standard deviations are reported for quantities on a linear scale, whereas geometric statistics are used for quantities evaluated on a logarithmic scale.

3.1. Steady-state simulation

3.1.1. Porous media flow setup

We consider a two-dimensional, laminar flow through a rectangular domain ($L_{x1} = 2.0\text{ m}$, $L_{x2} = 1.0\text{ m}$) containing randomly distributed circular obstacles of varying sizes. Obstacle surfaces are treated as no-slip walls. Periodic boundary conditions are imposed on the lateral boundaries, while the inlet prescribes a uniform velocity $U_\infty = 0.5\text{ m s}^{-1}$ and the outlet is maintained at fixed pressure. Simulations are performed using a single-precision D2Q9 SRT lattice Boltzmann method

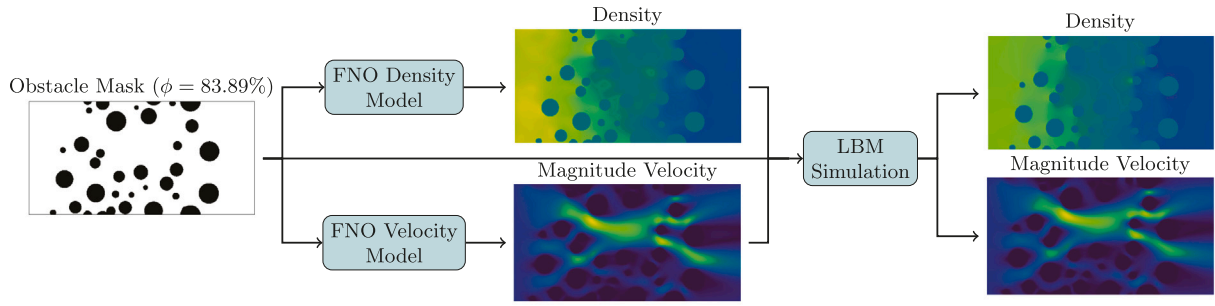


Fig. 4. Exemplary porous media initialization based on the workflow introduced in Fig. 3(a). From left to right, the obstacle mask is created and passed to the FNO models to predict density and velocity fields. The results are transferred to the LBM simulation loop that finally generates the steady-state fields (far right).

Table 1

Grid-refinement study for the porous-media configuration. Reported are masked L_2 errors between successive grid levels, relative L_2 errors, observed convergence orders, and Grid Convergence Index (GCI) values.

Variable	$e_{N \rightarrow 2N}$	$e_{2N \rightarrow 4N}$	Rel. e_{12} (%)	Rel. e_{23} (%)	p	GCI (%)
u	6.97×10^{-3}	2.54×10^{-3}	0.958	0.352	1.46	0.182
v	5.77×10^{-3}	1.95×10^{-3}	2.207	0.755	1.56	0.125
ρ	1.86×10^{-4}	4.40×10^{-5}	0.0185	0.0044	2.08	0.0017
p	7.44×10^{-2}	1.76×10^{-2}	0.0185	0.0044	2.08	0.680

with kinematic viscosity $\nu = 0.01 \text{ m}^2 \text{ s}^{-1}$. The porosity ranges from $\phi = 0.75$ to $\phi = 0.90$ and is defined as

$$\phi = \frac{A_{\text{pores}}}{A_{\text{total}}} = \frac{A_{\text{total}} - A_{\text{solid}}}{A_{\text{total}}}. \quad (11)$$

This range is representative of porous-media configurations considered in previous LBM porous media studies [39,40]. A summary of all simulation parameters is provided in Table 10.

Two FNO surrogates map the binary obstacle mask onto the corresponding steady-state density and velocity fields. All FNO models are trained and validated using data generated on the baseline grid resolution $N = 512 \times 256$. An overview of the workflow is shown in Fig. 4. The FNO hyperparameters for the density and velocity models, together with the dataset parameters, are summarized in Tables 11–13.

To assess spatial discretization errors and quantify grid sensitivity, a grid-refinement study was conducted using three successively refined meshes. The baseline grid is denoted by $N = 512 \times 256$, while the refined grids contain $2N = 1024 \times 512$ and $4N = 2048 \times 1024$ cells, respectively, corresponding to a refinement ratio of $r = 2$ in each spatial direction. The geometry, physical parameters, and boundary conditions were kept identical across all simulations. Since the flow reaches a steady-state, convergence was evaluated using time-averaged solution fields over the final $10\Delta T$ (10 outputs) of each simulation. Errors were computed only within the fluid domain by applying a mask that excludes solid cells and an additional one-cell erosion layer adjacent to obstacle boundaries. This procedure reduces the influence of geometric interpolation artifacts arising from the staircase representation of curved boundaries. The observed order of convergence was estimated from masked L_2 -norm differences between successive grid levels.

Table 1 shows monotonic convergence for all investigated quantities. The density and pressure fields exhibit approximately second-order convergence, with an observed convergence order of $p = 2.08$. This behavior indicates that the bulk flow quantities are only moderately affected by the geometric discretization errors introduced by the porous geometry. Since pressure is not directly solved by the lattice Boltzmann solver but obtained from density through the equation of state, the pressure and density fields exhibit identical relative errors and convergence rates. The larger absolute pressure errors arise solely from differences in scaling.

The velocity components exhibit lower convergence rates of $p = 1.46$ and $p = 1.56$ for u and v , respectively. Such behavior is commonly observed in porous-media simulations due to geometric discretization errors introduced by the staircase representation of curved obstacles and the bounce-back treatment of no-slip boundaries. Nevertheless, both velocity components converge monotonically under grid refinement, with relative differences below 2.3% between the N and $2N$ grids and below 0.8% between the $2N$ and $4N$ grids.

To further quantify the remaining discretization uncertainty, the Grid Convergence Index (GCI) was evaluated using the two finest grid levels. All reported GCI values remain below 1%, with values below 0.2% for the velocity components and below 0.01% for density. These results indicate that the uncertainty associated with spatial discretization is small relative to the magnitude of the solution variables. Moreover, the monotonic error reduction and the consistency of the estimated convergence orders suggest that the simulations are approaching the asymptotic convergence regime.

Although the $2N$ grid provides improved velocity-field accuracy, the baseline grid captures the dominant flow features while reducing computational cost by approximately a factor of four. It was therefore adopted as the baseline resolution for all training and validation datasets.

For the porous-media super-resolution experiments, the $2N$ grid was selected as the high-resolution reference solution. The grid-convergence results indicate that the $2N$ solution reduces the discretization error relative to the baseline N grid while remaining computationally feasible, particularly for the velocity field, where relative errors of approximately 1%–2% are observed. Consequently, the $N \rightarrow 2N$ mapping constitutes a meaningful super-resolution task in which the higher-resolution simulations contain additional physically relevant information beyond the coarse-grid solution. Importantly, the FNO models were trained and validated exclusively on data generated at resolution N . The $2N$ simulations were never used during training and are used exclusively during inference as an unseen spatial resolution. Evaluation on the $2N$ grid therefore assesses the ability of the learned operators to transfer across spatial resolutions that were not observed during training, in addition to preserving predictive accuracy. The $4N$ simulations were employed exclusively for verification purposes within the grid-refinement study and were not used during model training or evaluation.

Since the estimated discretization errors at resolution N are on the order of 10^{-3} to 10^{-2} in absolute terms and below a few percent in relative terms, all simulations were iterated to convergence tolerances significantly smaller than the estimated spatial discretization error. Therefore, the reported results are dominated by the accuracy of the spatial discretization rather than by incomplete iterative convergence.

3.1.2. Porous media flow results

Convergence of the velocity components, density field, and pressure drop relative to a steady-state reference solution is shown in Figs. 5(a) to 5(d). The reference is defined as the temporal average over the final

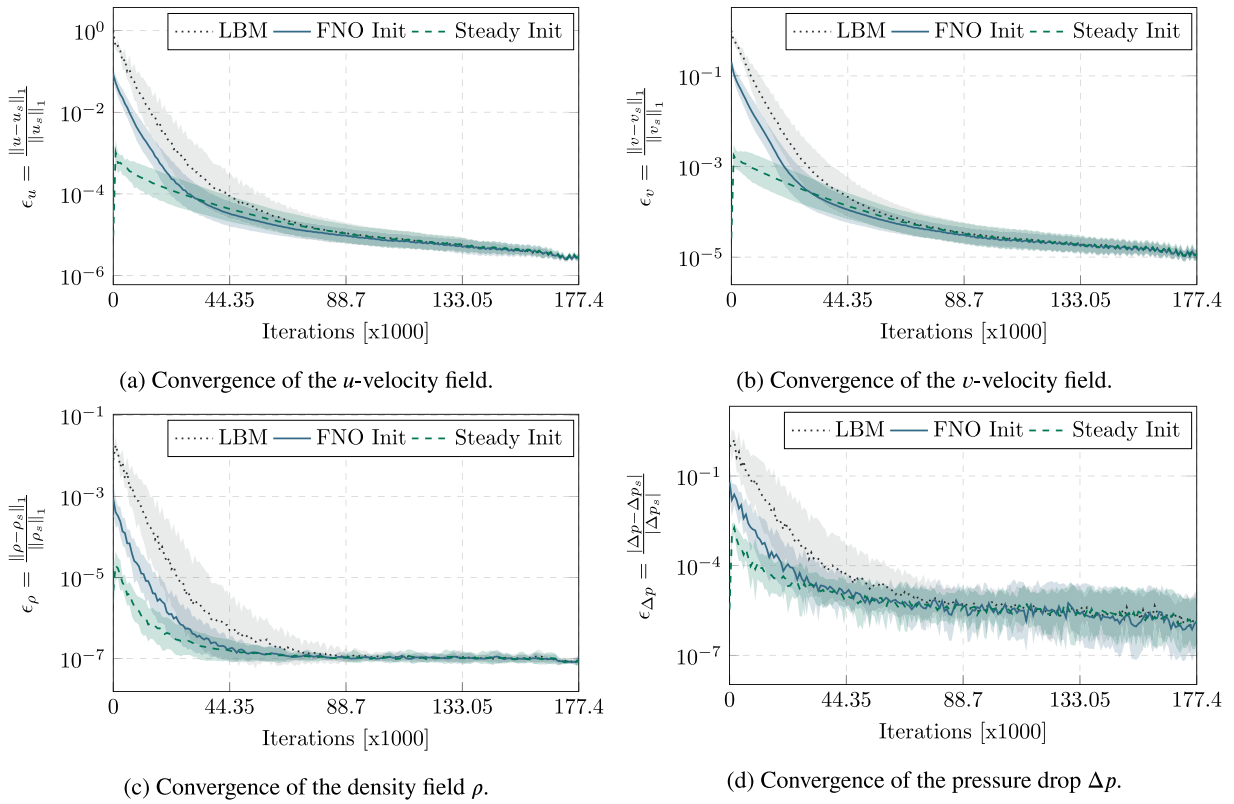


Fig. 5. Convergence of the velocity components u and v , density field ρ , and pressure drop Δp , with respect to the steady-state reference solution. The steady-state is computed as the average of the last 10 simulation outputs ($10\Delta T$). The pressure drop is defined as the difference between inflow and outflow pressures, and its error is computed relative to the steady-state pressure drop. Geometric mean and standard deviation ($\pm 1\sigma$, shaded bands) across 20 trajectories are depicted. For reference, the LBM is additionally initialized with the numerical steady-state solution and run until convergence (“Steady Init”).

$10\Delta T$ (10 outputs) of each trajectory. Errors are reported as relative L_1 norms for u , v , and ρ , and as the relative error of the pressure drop Δp between inlet and outlet. Shaded bands denote the standard deviation across 20 trajectories.

Initializing the LBM simulation with FNO-predicted fields accelerates convergence for all investigated quantities relative to the conventional initialization ($\rho = 1$, $u = v = 0$), while converging to the same steady-state solution. As shown in Table 2, the largest runtime reductions at $\epsilon_{\text{tol}} = 10^{-3}$ are obtained for density (87.4%) and pressure drop (55.4%), whereas the velocity components exhibit more moderate improvements of 29.6% for u and 22.4% for v . At stricter tolerances, the relative advantage decreases but remains positive. For density, the FNO initialization reduces the time required to reach the single-precision limit ($\epsilon_{\text{tol}} = 2 \times 10^{-7}$) by approximately 13.6% compared with conventional initialization.

Overall, these results demonstrate that FNO-based initialization provides a consistent and quantifiable acceleration of steady-state LBM convergence without compromising accuracy.

3.1.3. Porous media flow discretization invariance study

In this experiment, the FNO models trained exclusively on data at resolution N are evaluated on simulations performed at resolution $2N$, thereby assessing transfer across previously unseen spatial discretizations. Convergence of the velocity components, density field, and pressure drop relative to a steady reference solution of $2N$ is shown in Figs. 6(a) to 6(d). The reference is defined as the temporal average over the final $10\Delta T$ (10 outputs) of each trajectory. Errors are reported as relative L_1 norms for u , v , and ρ , and as the relative error of the pressure drop Δp between inlet and outlet. Shaded bands denote the standard deviation across 10 trajectories.

Initializing the LBM with FNO-predicted fields accelerates convergence in all quantities compared with conventional initialization ($\rho = 1$,

Table 2

Average runtime reduction over 20 trajectories for velocity, density and pressure drop fields at different tolerances ϵ_{tol} . The runtime reduction was computed as $SU_{\text{FNO}} = (T_{\text{LBM}} - T_{\text{FNO}})/T_{\text{LBM}}$ for FNO initialized and $SU_{\text{SS}} = (T_{\text{LBM}} - T_{\text{SS}})/T_{\text{LBM}}$ for steady-state initialized LBM, respectively, with reference to conventionally initialized ($\rho = 1$, $u = v = 0$) LBM simulations. Reported values correspond to the mean runtime reduction across trajectories, with 95% confidence intervals computed from the distribution of per-trajectory runtime reduction values as $\bar{S} \pm 1.96\sigma_S/\sqrt{n}$.

ϵ_{tol}	Quantity	SU_{FNO} (%)	SU_{SS} (%)
1×10^{-3}	u	29.6[22.3, 36.9]	77.8 [66.4, 89.2]
	v	22.4[15.2, 29.7]	42.8 [31.7, 53.9]
	ρ	87.4[79.7, 95.1]	89.3 [87.3, 91.2]
	Δp	55.4[44.6, 66.3]	87.2 [82.9, 91.5]
1×10^{-4}	u	15.6[6.1, 25.0]	10.7 [−0.9, 22.2]
	v	6.43[−2.8, 15.7]	0.0 [−12.4, 12.4]
	ρ	63.0[54.2, 71.7]	93.02 [90.9, 95.1]
	Δp	41.9[34.2, 49.6]	77.8 [73.15, 82.5]
1×10^{-5}	ρ	52.5[45.4, 59.6]	84.9 [81.1, 88.7]
	Δp	13.9[−6.9, 34.6]	33.0 [5.0, 61.0]
1×10^{-6}	ρ	28.0[17.5, 38.53]	52.0 [38.0, 66.0]
2×10^{-7}	ρ	13.6[2.3, 24.9]	17.0 [5.7, 28.2]

$u = v = 0$), while all simulations ultimately reach the same steady-state error. As summarized in Table 3, speed-ups at $\epsilon_{\text{tol}} = 10^{-3}$ reach 93.6% for ρ , 59.5% for Δp , 54.1% for u and 44.5% for v . Both velocity and density fields reach $\epsilon_{\text{tol}} = 10^{-4}$ more than 50% faster when initialized with the FNO predictions. For ρ , the FNO initialization enables the solver to reach the single-precision limit ($\epsilon_{\text{tol}} = 2 \times 10^{-7}$) approximately 48.3% faster than the conventional LBM approach.

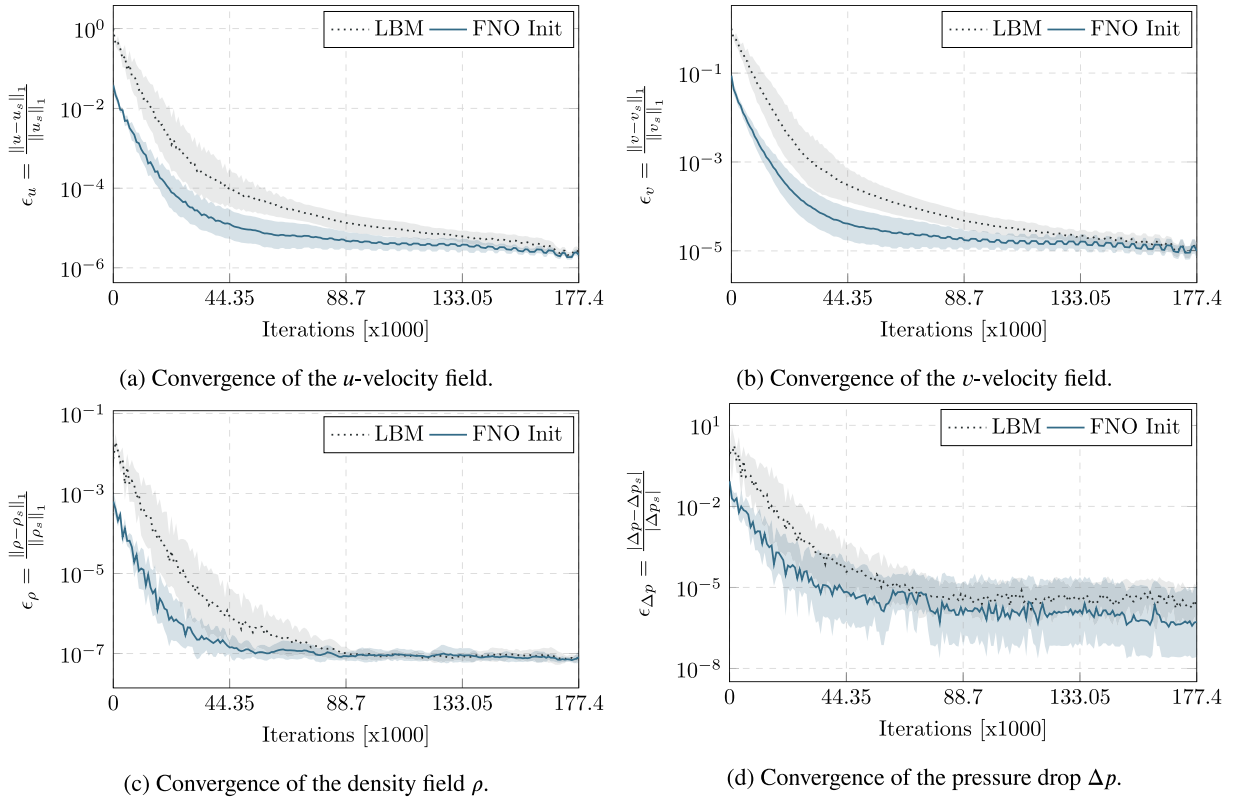


Fig. 6. Convergence of the velocity components u and v , density field ρ , and pressure drop Δp , with respect to the steady-state reference solution on a $2N$ resolution. The steady-state is computed as the average of the last 10 simulation outputs ($10\Delta T$). The pressure drop is defined as the difference between inflow and outflow pressures, and its error is computed relative to the steady-state pressure drop. Geometric mean and standard deviation ($\pm 1\sigma$, shaded bands) across 10 trajectories are depicted.

Table 3

Average runtime reduction over 10 trajectories for velocity, density and pressure drop fields at different tolerances ϵ_{tol} . The runtime reduction was computed as $SU_{\text{FNO}} = (T_{\text{LBM}} - T_{\text{FNO}})/T_{\text{LBM}}$ with reference to conventionally initialized ($\rho = 1, u = v = 0$) LBM simulations at $2N$ resolution. Reported values correspond to the mean runtime reduction across trajectories, with 95% confidence intervals computed from the distribution of per-trajectory runtime reduction values as $\bar{S} \pm 1.96\sigma_S/\sqrt{n}$.

	ϵ_{tol}			
Quantity	1×10^{-3}	1×10^{-4}	1×10^{-5}	2×10^{-7}
u	54.1 [47.7, 60.4]	51.8 [38.3, 65.3]	52.4 [38.8, 66.6]	–
v	44.5 [33.4, 55.6]	53.1 [39.7, 66.4]	–	–
ρ	93.6 [87.7, 99.6]	71.0 [65.1, 76.8]	53.1 [45.9, 60.2]	48.3 [33.0, 63.6]
Δp	59.5 [51.6, 67.4]	41.3 [31.3, 51.3]	20.5 [3.3, 37.7]	–

Compared with the training-resolution case, the super-resolution setting exhibits consistently larger convergence accelerations across all reported quantities. These results indicate that the learned operators retain predictive capability beyond the spatial resolution observed during training. Despite being trained exclusively on simulations at resolution N , the FNO-generated initial conditions accelerate convergence on the finer $2N$ grid across all investigated quantities. Overall, FNO-based initialization delivers a robust acceleration of steady-state LBM convergence without affecting the final converged solution, making it a promising approach for large-scale parameter studies and many-query porous media applications.

3.2. Unsteady simulation

3.2.1. Double shear layer setup

We consider the two-dimensional double shear layer in a periodic square domain ($L_{x_1} = L_{x_2} = 1.0\text{m}$). The initial horizontal velocity field

is prescribed by a piecewise hyperbolic-tangent profile with randomized shear-layer locations, while the vertical velocity introduces a small sinusoidal perturbation to trigger the instability as described by

$$u(x_1, x_2) = U \cdot \begin{cases} \tanh(A_1(x'_2 - x_2)), & y \geq 0.5 \\ \tanh(A_2(x_2 - x'_2)), & y < 0.5, \end{cases} \quad (12)$$

$$v(x_1, x_2) = 0.1 \cdot \sin\left(2\pi\left(x_1 + \frac{1}{4}\right)\right). \quad (13)$$

The characteristic velocity is set to $U = 1.0\text{m s}^{-1}$ and the kinematic viscosity to $\nu = 0.001\text{m}^2\text{s}^{-1}$. Based on the shear-layer thickness $\delta = 1/A$, with $A_1, A_2 \in [20, 120]$, the corresponding shear-layer Reynolds numbers range from approximately 8 to 50. The convective Reynolds number remains fixed at $Re = UL/\nu = 1000$.

Two groups of surrogate models were trained for this benchmark, each consisting of a density model and a velocity model. The first group contains a large 11.2M-parameter FNO, while the second group consists of a compact 2.6M-parameter FNO (see Table 4). Unless otherwise stated, all FNO models are trained and validated exclusively on simulations generated at the baseline resolution $N = 128 \times 128$. The large model remains stable during fully autoregressive rollout over the considered prediction horizon, whereas the compact model exhibits error accumulation and eventual destabilization. Hybrid FNO–LBM simulations follow the framework introduced in Section 2.3, where the FNO replaces a prescribed fraction of LBM time steps according to fixed coupling cadences (1:1, 1:2, and 1:3). A comparison of pure LBM, pure FNO, and hybrid rollout is shown in Fig. 7.

To quantify spatial discretization errors and assess grid sensitivity, a grid-convergence study was conducted using three successively refined meshes: $N = 128 \times 128$, $2N = 256 \times 256$, and $4N = 512 \times 512$. All simulations employed a refinement ratio of $r = 2$. The velocity components and density field were compared using timestep-wise

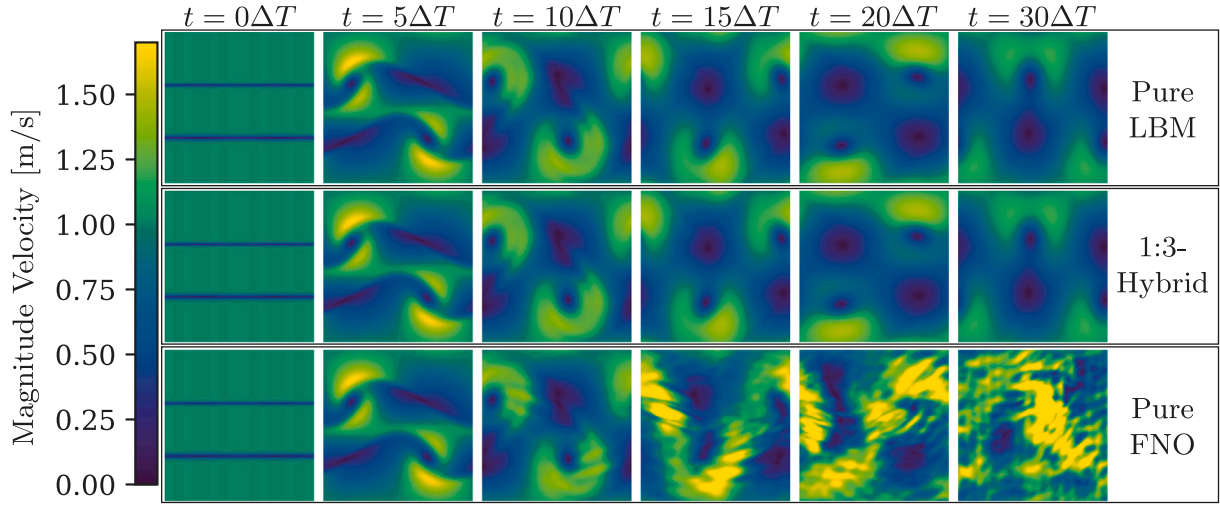


Fig. 7. Rollout realization with the 2.6M FNO model based on the workflow introduced in Fig. 3(b). The first row depicts a pure LBM simulation, in the middle is a hybrid with one quarter of the simulation steps replaced by the FNO and the bottom row shows the pure autoregressive rollout of the 2.6M FNO.

Table 4

Total parameters of the FNO models for the double shear prediction.

Group Name	Density Model	Velocity Model	Combined
11.2M FNO	5 621 835	5 621 875	11 243 710
2.6M FNO	151 811	2 409 771	2 561 582

Table 5

Summary of the grid convergence study using refinement ratio $r = 2$. Relative errors are reported in percent.

Field	$\bar{L}_{2,N,2N}$	$\bar{L}_{2,N,4N}$	Rel. $\bar{L}_{2,N,2N}$ (%)	Order p	Mean GCI (%)	Max GCI (%)
u	$1.38e^{-3}$	$3.25e^{-4}$	0.195	2.03	0.013	0.029
v	$1.23e^{-3}$	$2.81e^{-4}$	0.253	2.00	0.011	0.025
ρ	$4.50e^{-5}$	$7.00e^{-5}$	0.0045	-0.44	0.025	0.073

L_2 norms after restricting finer-grid solutions to the corresponding coarser resolution. Observed convergence orders were estimated using Richardson extrapolation.

As summarized in Table 5, both velocity components exhibit near-ideal second-order convergence. The observed convergence orders are $p = 2.03$ for u and $p = 2.00$ for v . Relative L_2 errors remain below 0.3% on the coarse-to-medium refinement level and decrease below 0.06% between the two finest grids. Refinement from $2N$ to $4N$ further reduced the relative errors to below 0.06%, indicating that only minor changes occur in the velocity solution as the grid is refined.

The density field yields an observed convergence order of $p = -0.44$. However, the corresponding relative errors remain below 0.01% throughout the grid study. For quantities whose inter-grid differences become extremely small relative to their magnitude, Richardson-based estimates may become highly sensitive to interpolation and numerical errors, rendering the observed order less reliable than for the velocity fields. Consequently, the negative value should not be interpreted as a lack of convergence, but rather as an indication that the density differences between grid levels are already extremely small.

Given the observed second-order convergence of the velocity fields, the sub-percent relative errors, and the low GCI values, the baseline grid $N = 128 \times 128$ was selected for dataset generation and all training-resolution experiments. The $2N$ grid serves as the high-resolution reference for the discretization-invariance study presented later in this section and is used exclusively during inference to assess transfer across previously unseen spatial resolutions.

3.2.2. Double shear layer results

We quantify accuracy with a normalized vorticity error and a mean-squared error across all fields, both computed per time step and as temporal averages over the $30\Delta T$ horizon (Figs. 8 and 9).

Across both model sizes, the global MSE decreases monotonically as the frequency of LBM corrections increases. The effect is moderate but consistent for the 11.2M model and substantially larger for the 2.6M model, for which hybrid coupling suppresses the instability observed in pure autoregressive rollout. Consequently, all hybrid configurations maintain bounded errors throughout the prediction horizon, with the 1:3-Hybrid configuration generally yielding the lowest global error (Figs. 8(b) and 8(d)). The time-resolved error curves corroborate this observation. Hybrid rollouts remain consistently below their pure-FNO counterparts and exhibit significantly reduced long-term error growth, demonstrating that periodic LBM corrections effectively stabilize surrogate predictions.

Vorticity errors exhibit the same trend. Global $\bar{\epsilon}_\omega$ decreases with increasing correction frequency, indicating improved preservation of small-scale rotational structures. The 2.6M pure FNO shows pronounced error growth over time, whereas all hybrids remain stable and attain vorticity errors comparable to the large-model hybrid, see Fig. 9.

Table 6 summarizes relative reduction in mean global error achieved by the hybrid models compared to their pure FNO counterparts. For the 11.2M model, hybridization yields global MSE reductions of 17.3%, 48.7%, and 60.0% for 1:1, 1:2, and 1:3 configurations, respectively, while vorticity reductions range from 19.8% to 42.1%. The impact is considerably larger for the compact 2.6M model. Relative MSE reductions exceed 99.9%, and vorticity errors decrease by approximately 99%. Hybrid coupling therefore enables the compact surrogate to operate within an error regime comparable to that of the large-model hybrid configurations. Overall, these results demonstrate that hybridization improves both accuracy and long-horizon stability, enabling substantially smaller surrogate models to achieve competitive performance.

The computational performance of the hybrid approach relative to pure LBM is summarized in Table 7. Wall-clock runtimes were averaged over the first six trajectories, with runtime variations below 0.8% of the corresponding mean. The attainable runtime reduction depends strongly on the ratio between the FNO prediction interval ΔT and the numerical time step Δt . Consequently, the reported values should be interpreted as problem-dependent rather than universal performance limits. In particular, the present study employs a comparatively conservative surrogate stride, with each FNO evaluation replacing only

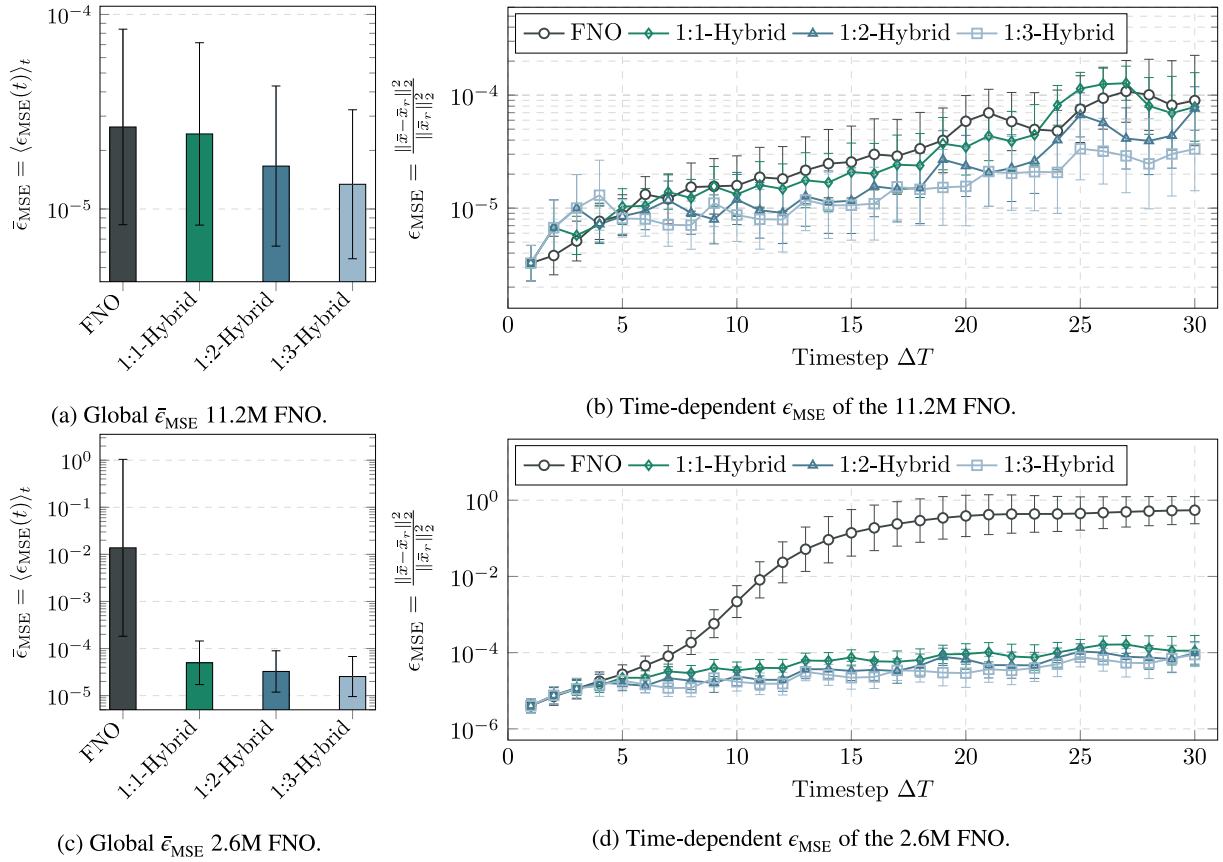


Fig. 8. Global ($\bar{\epsilon}_{\text{MSE}}$) and time-dependent (ϵ_{MSE}) MSE of the 11.2M and 2.6M FNO models in pure and hybrid rollout scenarios with respect to the reference pure LBM simulation. Geometric mean and standard deviation ($\pm 1\sigma$) across 100 trajectories are depicted.

Table 6

Relative reduction of the mean global error over 100 trajectories for hybrid FNO–LBM models. Error reduction is computed as $(\bar{\epsilon}_{\text{FNO}} - \bar{\epsilon}_{\text{Hybrid}})/\bar{\epsilon}_{\text{FNO}}$. Reported values are means with corresponding 95% confidence intervals, obtained via error propagation (delta method).

Metric	Model	1:1-Hybrid (%)	1:2-Hybrid (%)	1:3-Hybrid (%)
$\bar{\epsilon}_{\omega}$	11.2M	19.8 [11.2, 28.4]	35.6 [28.1, 43.0]	42.1 [35.0, 49.2]
	2.6M	98.7 [98.4, 98.9]	99.0 [98.8, 99.2]	99.1 [98.9, 99.3]
$\bar{\epsilon}_{\text{MSE}}$	11.2M	17.34 [-14.7, 49.4]	48.7 [28.8, 68.6]	60.0 [44.1, 76.0]
	2.6M	99.97 [99.96, 99.98]	99.98 [99.97, 99.99]	99.99 [99.98, 99.99]

223 underlying LBM time steps. The resulting speed-ups therefore correspond to a lower-bound estimate of the acceleration potential of the hybrid framework rather than a fundamental limit. For problems requiring finer temporal resolutions and thus smaller simulation time steps, the same physical prediction horizon would correspond to substantially larger values of $\alpha = \Delta T_{\text{FNO}}/\Delta t_{\text{LBM}}$, allowing each surrogate evaluation to bypass a greater number of LBM updates and thereby yielding larger runtime reductions. Both surrogate models achieve comparable runtime reductions despite their different parameter counts, indicating that the overall runtime reduction is primarily governed by the prescribed coupling cadence. Depending on the hybrid configuration, runtime reductions range from approximately 5% to 12%.

We note that in many practical deployment scenarios, the required training data is already available from prior simulations, such as parameter studies or design-space exploration. In such cases, the cost of data generation is negligible, and the surrogate primarily serves to accelerate repeated evaluations. Consequently, the reported runtime reductions translate directly into practical computational savings without

Table 7

We compute the relative reduction of simulation runtime $\epsilon_T = (T_{\text{LBM}} - T_{\text{Hybrid}})/T_{\text{LBM}}$ with corresponding 95% confidence intervals for runtime reduction using a paired non-parametric bootstrap with 100,000 resamples and percentile-based bounds. As additional reference, the relative reduction of simulation steps $\epsilon_N = (N_{\text{LBM}} - N_{\text{Hybrid}})/N_{\text{LBM}}$ for each hybrid approach is given compared to pure LBM simulation.

Metric	Model	1:1-Hybrid (%)	1:2-Hybrid (%)	1:3-Hybrid (%)
ϵ_T	11.2M FNO	11.1 [10.4, 11.7]	7.7 [7.1, 8.2]	5.1 [4.6, 5.6]
	2.6M FNO	11.8 [11.4, 12.5]	7.7 [7.2, 8.2]	5.7 [5.1, 6.2]
ϵ_N	—	50	34	25

requiring extensive amortization of training cost. Taken together, these results demonstrate that hybrid FNO–LBM models achieve a favorable balance between accuracy, stability, and efficiency, delivering consistent runtime savings while preserving solution fidelity, particularly when applied to compact surrogate architectures.

3.2.3. Double shear layer discretization invariance study

To assess discretization invariance, hybrid simulations were evaluated with respect to purely numerical simulations on a grid with twice the training resolution ($2N = 256 \times 256$) using ten previously unseen double-shear-layer configurations. Representative rollouts are shown in Fig. 10. Both FNO models exhibit visible aliasing artifacts when deployed autoregressively in a zero-shot resolution-transfer setting. Periodic LBM corrections attenuate these artifacts but do not eliminate them entirely, particularly for the 11.2M model. These observations are consistent with previous studies on neural operators under resolution extrapolation, which reported increased aliasing and more rapid

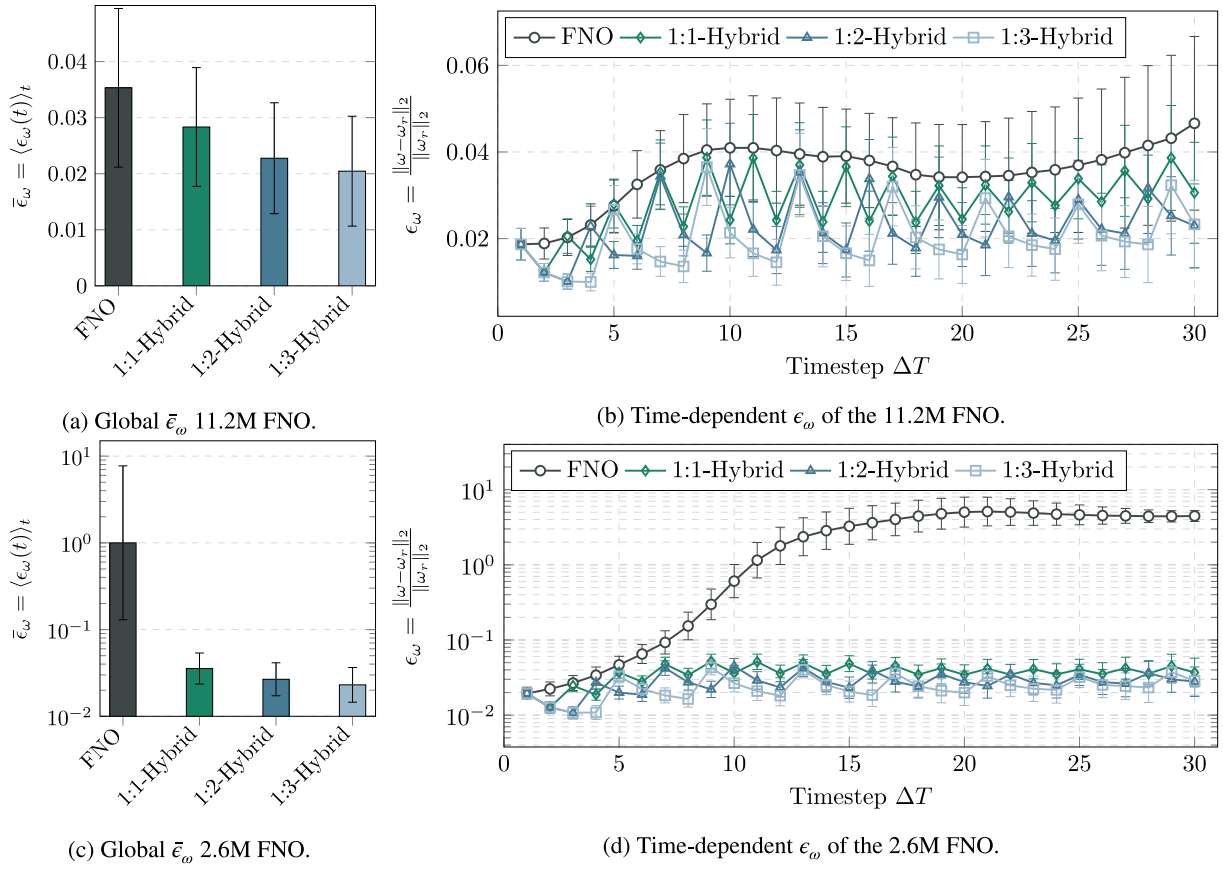


Fig. 9. Global ($\bar{\epsilon}_\omega$) and time-dependent (ϵ_ω) relative L_2 vorticity error of the 11.2M and 2.6M FNO models in pure and hybrid rollout scenarios with respect to the reference pure LBM simulation. Arithmetic mean and standard deviation ($\pm 1\sigma$) are reported for linear-scale quantities, geometric means and standard deviations ($\pm 1\sigma$) for logarithmic quantities across 100 trajectories.

error accumulation when models are evaluated outside their training discretization [41].

To quantify predictive accuracy, we compute the mean-squared error (MSE) across all flow fields, both at individual time steps and as a temporal average over the rollout horizon (Fig. 11). Across both surrogate model sizes, the global MSE decreases as the frequency of LBM corrections increases (1:1 $\rightarrow$ 1:2 $\rightarrow$ 1:3) relative to pure autoregressive FNO rollouts. For the 11.2M FNO, the improvement is modest but consistent, yielding an average relative error reduction of 4.7%–40.4% depending on the hybrid cadence. In contrast, the 2.6M FNO benefits substantially more from hybrid coupling, with MSE reductions ranging from 66.0% to 83.4%, see Table 8.

Although hybrid coupling reduces prediction errors relative to pure autoregressive rollout, both surrogate models experience a substantial loss of accuracy at twice the training resolution. From the first prediction step onward, MSE values increase from the training-resolution regime of approximately $\mathcal{O}(10^{-6})$ to approximately $\mathcal{O}(10^{-3})$, corresponding to an error increase of roughly three orders of magnitude (Figs. 11(b) and 11(d)). The hybrid framework successfully suppresses a portion of the aliasing artifacts and preserves the dominant flow structures (Fig. 10). However, once surrogate prediction errors reach $\mathcal{O}(10^{-3})$, periodic numerical corrections alone are insufficient to recover the solution fidelity observed at the training resolution. Achieving comparable accuracy would require more frequent LBM corrections, which would proportionally reduce the attainable computational savings.

Overall, the attainable runtime reduction in the super-resolution setting is comparable to that observed at the training resolution, ranging from 4.4% to 12.0% depending on the hybrid coupling cadence (Table 9).

Table 8

Relative reduction of the mean global error over 10 trajectories for hybrid FNO–LBM models. Error reduction is computed as $(\bar{\epsilon}_{\text{FNO}} - \bar{\epsilon}_{\text{Hybrid}})/\bar{\epsilon}_{\text{FNO}}$. Reported values are means with corresponding 95% confidence intervals, obtained via error propagation (delta method).

Metric	Model	1:1-Hybrid (%)	1:2-Hybrid (%)	1:3-Hybrid (%)
$\bar{\epsilon}_{\text{MSE}}$	11.2M	4.7 [−19.1, 28.5]	40.4 [24.4, 56.4]	36.7 [21.3, 52.0]
	2.6M	66.0 [57.5, 74.4]	78.6 [73.2, 83.9]	83.4 [79.2, 87.6]

Table 9

We compute the relative reduction of simulation runtime $\epsilon_T = (T_{\text{LBM}} - T_{\text{Hybrid}})/T_{\text{LBM}}$ with corresponding 95% confidence intervals for runtime reduction using a paired non-parametric bootstrap with 100,000 resamples and percentile-based bounds over 10 trajectories. As additional reference, the relative reduction of simulation steps $\epsilon_N = (N_{\text{LBM}} - N_{\text{Hybrid}})/N_{\text{LBM}}$ for each hybrid approach is given compared to pure $2N$ LBM simulation.

Metric	Model	1:1-Hybrid (%)	1:2-Hybrid (%)	1:3-Hybrid (%)
ϵ_T	11.2M FNO	10.5 [9.2, 11.7]	9.8 [8.5, 10.9]	4.4 [2.9, 5.9]
	2.6M FNO	12.0 [10.3, 13.5]	9.1 [7.7, 10.2]	4.1 [1.9, 6.4]
ϵ_N	–	50	34	25

In summary, the discretization-invariance study shows that the investigated FNO surrogates do not generalize reliably beyond their training resolution when applied autoregressively. Hybrid FNO–LBM coupling consistently improves accuracy and mitigates aliasing artifacts relative to pure autoregressive rollout, but cannot fully compensate for the degradation associated with autoregressive zero-shot super-resolution. These results suggest that the dominant limitation in the present setting arises from the surrogate models rather than from the

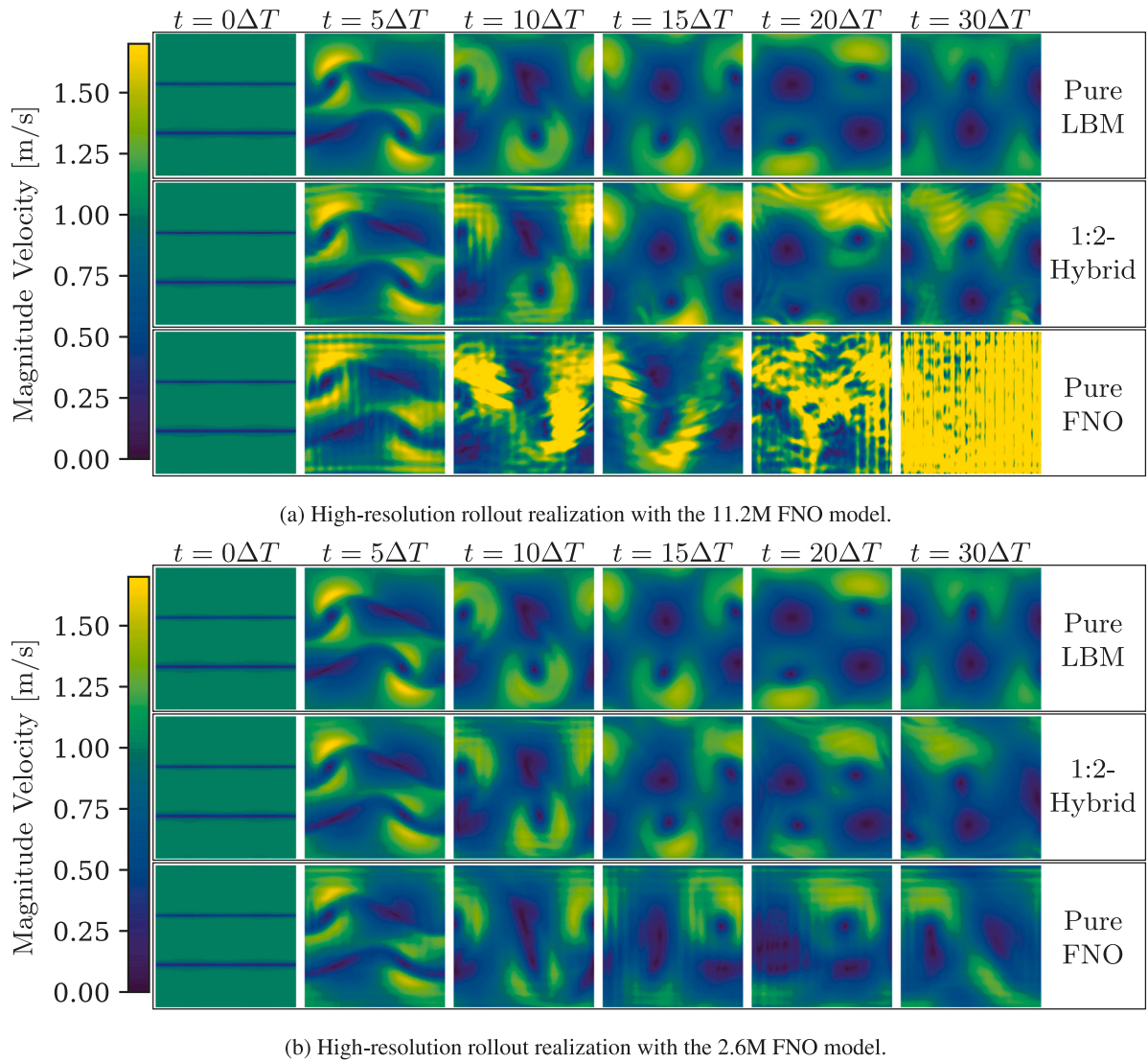


Fig. 10. High-resolution rollout realizations with both 11.2M and 2.6M FNO model based on the workflow introduced in Fig. 3(b). The first row depicts a pure LBM simulation, in the middle is a hybrid with one third of the simulation steps replaced by the FNO and the bottom row shows the pure autoregressive rollout of the respective FNO.

hybrid coupling strategy itself. While hybridization remains effective for stabilizing predictions and providing runtime savings, achieving robust acceleration across unseen discretizations will likely require either explicit multi-resolution training or surrogate architectures with stronger discretization-invariant properties.

4. Conclusion

This work investigated the coupling of Fourier Neural Operator (FNO) surrogates with lattice Boltzmann method (LBM) solvers for accelerating both steady-state and unsteady two-dimensional, single-phase, low-Reynolds-number flows while preserving physical accuracy. Two representative benchmark problems were considered: a steady porous-media flow and an unsteady double shear layer, allowing the proposed framework to be evaluated in both iterative steady-state and time-dependent prediction settings.

For steady porous-media flows, FNO-based initialization substantially reduced the time required for the LBM solver to reach the converged solution while preserving the final steady-state accuracy. The largest benefits were observed for density and pressure-drop convergence, demonstrating that surrogate predictions can provide effective

physically informed initial conditions for iterative CFD simulations. Notably, these benefits persisted when the models were applied at twice the training resolution. Under resolution transfer to the $2N$ grid, FNO-initialized simulations achieved even larger runtime reductions than those observed at the training resolution while converging to the same steady-state solution. This result indicates that the surrogate retains dominant large-scale flow structures beyond its training discretization and can therefore serve as an effective accelerator for higher-resolution steady-state simulations.

For the unsteady double shear layer, hybrid FNO-LBM coupling consistently improved predictive accuracy and long-horizon stability relative to purely autoregressive surrogate rollouts. Periodic numerical corrections effectively limited error accumulation and enabled a compact 2.6M-parameter FNO, which is unstable in standalone inference, to operate within an accuracy regime comparable to that of a substantially larger 11.2M-parameter model. Hybrid coupling reduced global MSE by up to 99.9% and vorticity error by up to 99.1%, demonstrating a strong regularizing effect of numerical correction. These findings show that hybridization can combine the computational efficiency of neural operators with the robustness of physics-based numerical solvers through periodic re-alignment with the underlying flow dynamics.

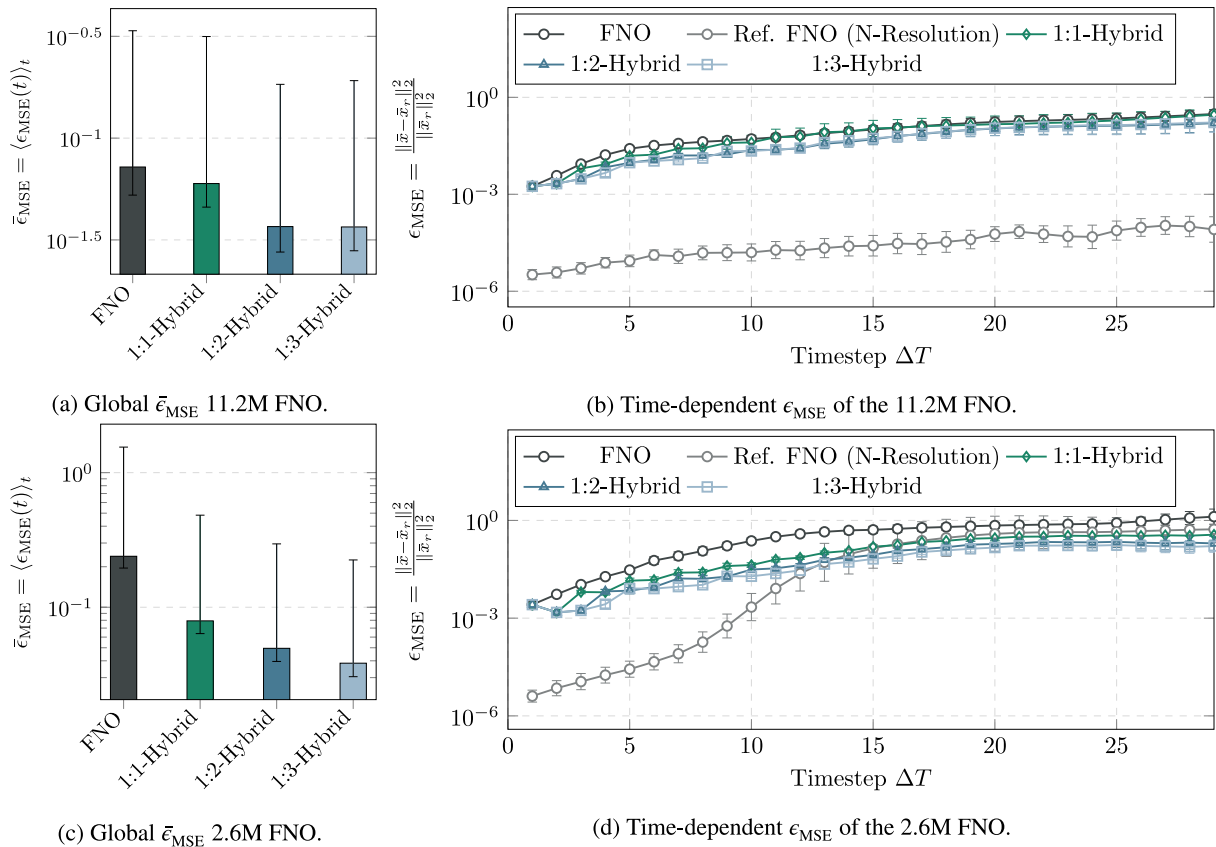


Fig. 11. Global ($\bar{\epsilon}_{\text{MSE}}$) and time-dependent (ϵ_{MSE}) MSE of the 11.2M and 2.6M FNO models in pure and hybrid rollout scenarios with respect to the reference pure LBM simulation for $2N$ -resolution rollout. Geometric mean and standard deviation ($\pm 1\sigma$) across 10 trajectories are depicted. For reference, we added the time-dependent (ϵ_{MSE}) MSE of the 11.2M and 2.6M FNO models for N -resolution rollout.

Although the primary focus of this work is on FNO–LBM coupling, an additional comparison with a residual U-Net surrogate yielded qualitatively similar behavior. Hybrid UNet–LBM simulations consistently reduced both mean-squared and vorticity errors relative to pure autoregressive U-Net rollouts, demonstrating that the benefits of periodic numerical correction are not restricted to neural operators. While the FNO retained a more favorable balance between predictive performance and computational efficiency, the U-Net results indicate that the proposed hybridization strategy is largely surrogate-model agnostic.

The discretization-invariance studies revealed a clear distinction between the steady and unsteady applications. For steady-state initialization, the FNO predictions generalized sufficiently well across resolutions to provide substantial convergence acceleration even on unseen finer grids. In contrast, the autoregressive double-shear-layer surrogates exhibited a pronounced loss of accuracy when evaluated beyond their training resolution. Although hybrid FNO–LBM coupling consistently reduced errors and attenuated aliasing artifacts, it could not recover the level of accuracy achieved at the training resolution. These results suggest a fundamental distinction between steady-state initialization and autoregressive prediction under resolution transfer. Steady-state initialization remains comparatively robust to discretization changes, whereas long-horizon autoregressive prediction is substantially more sensitive to unseen spatial resolutions.

Overall, the proposed framework demonstrates that machine-learning surrogates, and FNOs in particular, can serve two complementary roles within CFD workflows: as physics-informed initializers for accelerating steady-state convergence and as predictive components within hybrid solver–surrogate simulations. The steady-state experiments show that surrogate-based initialization can remain effective beyond the training resolution, whereas the unsteady experiments show that hybrid numerical corrections substantially improve stability and

accuracy but cannot fully compensate for poor autoregressive zero-shot resolution transfer. Together, these findings highlight both the practical potential of hybrid ML–CFD workflows and the importance of developing neural operators with stronger discretization-aware generalization properties. More generally, the results demonstrate that the frequency of numerical correction constitutes an effective control parameter for balancing runtime reduction against predictive accuracy in hybrid ML–CFD simulations.

Important extensions remain to be explored, including higher Reynolds number regimes requiring different collision models, three-dimensional configurations, complex boundary conditions, adaptive hybrid switching criteria and autoregressive discretization-invariant machine learning architectures. While the proposed coupling operates on macroscopic fields and is therefore, in principle, agnostic to the specific LBM formulation, the additional U-Net experiments further suggest that the framework is not restricted to a particular surrogate architecture. Nevertheless, the performance of the coupled system depends strongly on the predictive accuracy, inference cost, and resolution-generalization properties of the chosen surrogate. The present work therefore represents a controlled proof-of-concept study, and further validation is required to assess the generality of the approach in more complex CFD settings.

CRediT authorship contribution statement

A. Junk: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **J.M. Winter:** Writing – review & editing, Supervision, Software, Formal analysis, Conceptualization. **M. Tütken:** Writing – review & editing, Data curation. **S. Schmidt:** Writing – review & editing, Supervision, Funding

Table 10

Porous media flow simulation parameters.

Boundary conditions	Physical parameters	LBM Algorithm
Inlet: Fixed Inflow	$U_{\infty} = 0.5 \frac{m}{s}$	Single Precision
Sides: Periodic	$\nu = 0.01 \frac{m^2}{s}$	D2Q9
Outlet: Fixed Pressure	$\phi = 0.75 - 0.9$	SRT

Table 11

Porous media FNO density model hyperparameters.

Decoder	FNO	Scheduler
Out Features: 1	In Channels: 1	Learning Rate: 1.e-2
Layers: 3	Latent Channels: 16	Decay Rate: 0.85
Layer Size: 32	Layers: 3	Decay Frequency: 10.Epoch
	Fourier Modes: 6	Type: LambdaLR
	Padding: 9	Optimizer AdamW
Total Model Parameters: 114 273; Time per Epoch: 25.9s		
Batch Size: 64,	Epochs: 300	

Table 12

Porous media FNO velocity model hyperparameters.

Decoder	FNO	Scheduler
Out Features: 2	In Channels: 1	Learning Rate: 1.e-2
Layers: 3	Latent Channels: 16	Decay Rate: 0.85
Layer Size: 64	Layers: 6	Decay Frequency: 10.Epoch
	Fourier Modes: 12	Type: LambdaLR
	Padding: 9	Optimizer AdamW
Total Model Parameters: 896 082; Time per Epoch: 89.8 s		
Batch Size: 64,	Epochs: 300	

acquisition, Conceptualization. **N.A. Adams:** Writing – review & editing, Supervision, Project administration, Conceptualization, Funding acquisition,.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Microsoft Copilot to improve readability and language. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. General machine learning setup

All models were trained using PhysicsNeMo version 0.8.0 on an NVIDIA RTX A4000 with initialization based on fixed seed number 42.

Appendix B. Porous media flow setup

See [Tables 10–13](#).

Appendix C. Double shear layer flow setup

See [Tables 14–16](#).

See [Fig. 12](#).

Table 13

Porous media dataset parameters.

Dataset	Sample Size	Resolution
Training Data	1700	512×256
Validation Data	200	512×256
Inference	20 Trajectories	512×256
Wall-Clock-Time per Simulation: 354 s		
High Resolution Test	10 Trajectories	1024×512
Wall-Clock-Time per Simulation: 1363 s		

Table 14

Double shear FNO density models hyperparameters.

Decoder	FNO	Scheduler
Out Features: 1	In Channels: 1	Learning Rate: 1.e-2
Layers: 1	Latent Channels: 14	Decay Rate: 0.85
Layer Size: 32	Layers: 3,7	Decay Frequency: 8.Epoch
	Fourier Modes: 8,32	Type: LambdaLR
	Padding: 9	Optimizer AdamW
Total Parameters (3 Layers, 8 Fourier Modes): 151 811		
Time per Epoch: 17.2 s		
Total Parameters (7 Layers, 32 Fourier Modes): 5 621 835		
Time per Epoch: 21.1 s		
Batch Size: 64,	Epochs: 300	

Table 15

Double shear FNO velocity models hyperparameters.

Decoder	FNO	Scheduler
Out Features: 2	In Channels: 2	Learning Rate: 1.e-2
Layers: 1	Latent Channels: 14	Decay Rate: 0.85
Layer Size: 32	Layers: 3,7	Decay Frequency: 8.Epoch
	Fourier Modes: 32	Type: LambdaLR
	Padding: 9	Optimizer AdamW
Total Model Parameters (3 Layers): 2 409 771; Time per Epoch: 18.9 s		
Total Model Parameters (7 Layers): 5 621 875; Time per Epoch: 21.4 s		
Batch Size: 64,	Epochs: 300	

Table 16

Double shear layer dataset parameters.

Dataset	Samples	Resolution
Training	3000 Pairs	128×128
Validation	1000 Pairs	128×128
Inference	100 Trajectories	128×128
Wall-Clock-Time per Simulation: 18 s		
High Resolution Test	10 Trajectories	256×256
Wall-Clock-Time per Simulation: 26 s		

Appendix D. UNet hybridization: Double shear layer results

To assess whether the observed performance of the proposed hybrid framework depends on the choice of surrogate model, we additionally evaluate a residual U-Net architecture. This experiment serves as a comparison against a conventional convolutional neural network architecture.

Preliminary experiments indicated that additional architectural modifications were required for the U-Net to attain one-step prediction errors comparable to those of the FNO. To improve predictive accuracy, two modifications were introduced. First, the network was trained to predict the temporal residual

$$\bar{x}_{t+1} = \bar{x}_t + \text{UNet}(\bar{x}_t), \quad (14)$$

instead of directly predicting the next state. Second, normalized Cartesian coordinate channels were concatenated to the input, analogous to positional encodings, providing explicit spatial information to the convolutional network. These modifications were applied for both density and velocity prediction.

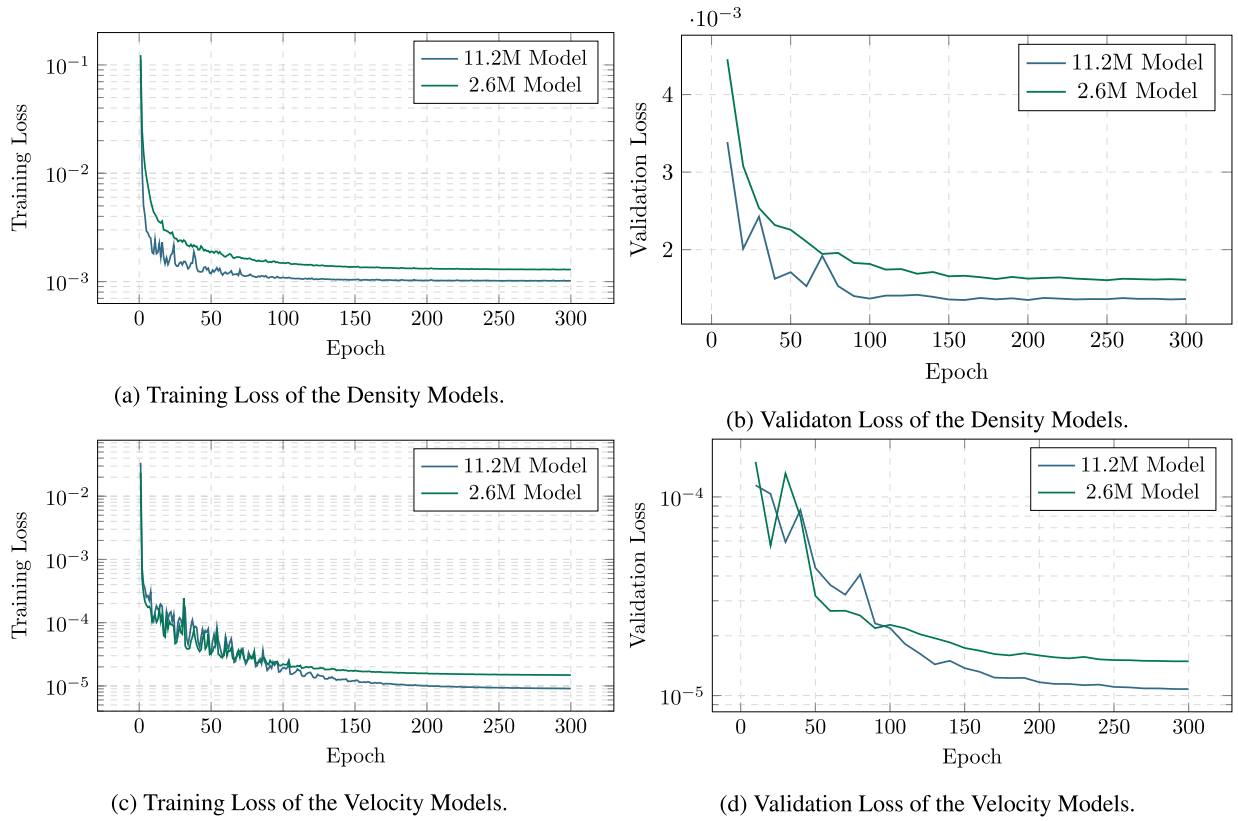


Fig. 12. Training and Validation Loss of all FNO Models trained on the Double Shear Layer.

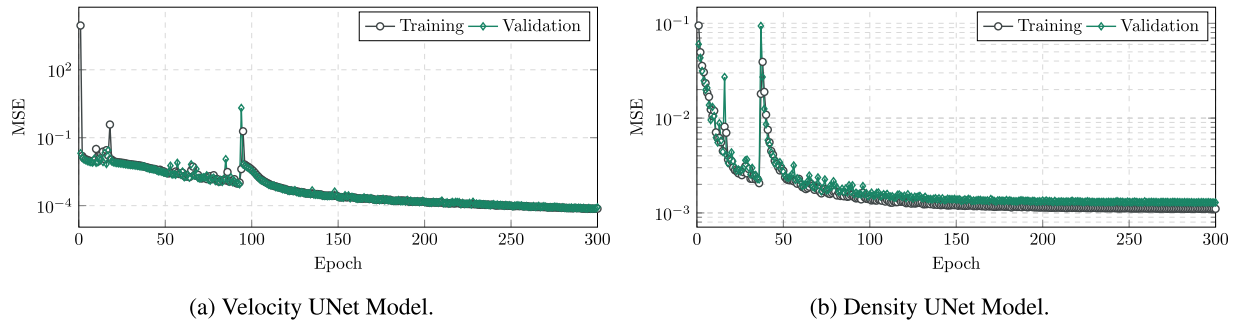


Fig. 13. Training and Validation Loss (MSE) for the UNet models for density and velocity prediction.

The resulting architecture is a residual U-Net. The encoder comprises four residual blocks with three downsampling operations, followed by a residual bottleneck and three upsampling decoder blocks. Each stage employs residual convolutional blocks with GELU activations and Group Normalization. Downsampling is realized using strided convolutions and upsampling through bilinear interpolation followed by convolutional processing. All convolutions employ circular padding to enforce consistency with the periodic boundary conditions of the flow domain.

The network contains approximately 225k trainable parameters for density prediction and 14.3M parameters for velocity prediction. Training was performed on the dataset described in Table 16. Apart from a learning-rate decay applied every tenth epoch, the training setup was identical to that used for the FNO experiments (Table 15). Training and validation losses are shown in Fig. 13.

We quantify accuracy with a relative vorticity error and a mean-squared error across all fields, both computed per timestep and as temporal averages over the $30\Delta T$ horizon (Figs. 14(b) and 14(d)), identical to the previously investigated FNO.

Both global and time-resolved error curves decrease when LBM corrections are applied (Fig. 14), similar to the trends observed for the FNO experiments, indicating that the benefits of hybridization are not specific to the FNO and generalize to a conventional convolutional neural network surrogate. Hybrid rollouts remain consistently below their pure-UNet counterparts and exhibit significantly reduced long-term error growth, demonstrating that periodic LBM corrections effectively correct errors introduced by surrogate predictions.

Table 17 summarizes relative reduction in mean global error achieved by the hybrid models compared to their pure UNet counterparts. Hybridization yields global MSE reductions of 92.6%, 97.1%, and 95.7% for 1:1, 1:2, and 1:3 configurations, respectively, while vorticity reductions range from 75.3% to 82.5%. Overall, these results demonstrate a significant accuracy gain through hybrid UNet-LBM coupling.

The computational performance of the hybrid approach relative to pure LBM is summarized in Table 18. The attainable runtime reduction again depends strongly on the ratio between the UNet prediction interval ΔT and the numerical time step Δt . Consequently, the reported

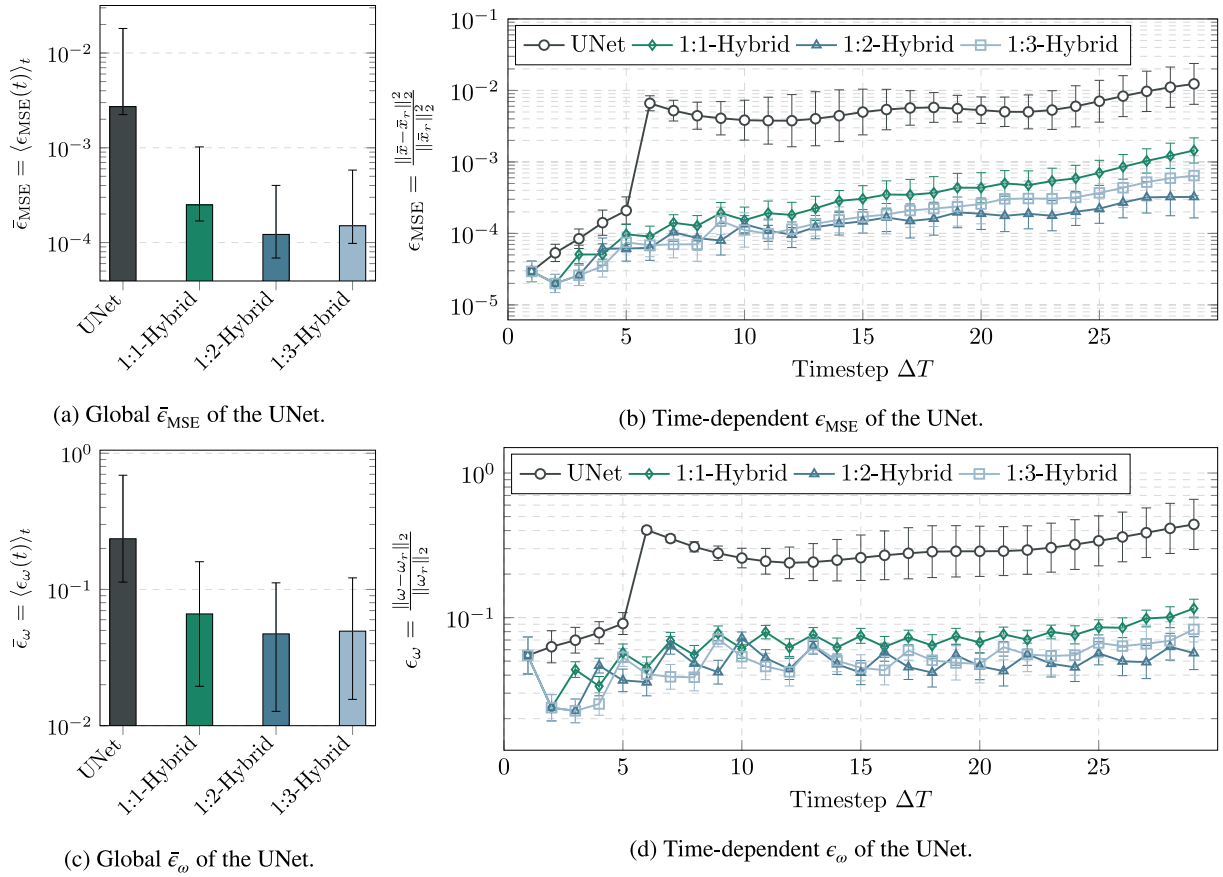


Fig. 14. Global ($\bar{\epsilon}_{\text{MSE}}$), time-dependent (ϵ_{MSE}) MSE, global ($\bar{\epsilon}_{\omega}$) and time-dependent (ϵ_{ω}) relative L_2 vorticity error of the UNet in pure and hybrid rollout scenarios with respect to the reference pure LBM simulation. Geometric mean and standard deviation ($\pm 1\sigma$) across 10 trajectories are depicted.

Table 17

Relative reduction of the mean global error over 10 trajectories for hybrid UNet–LBM models. Error reduction is computed as $(\bar{\epsilon}_{\text{UNet}} - \bar{\epsilon}_{\text{Hybrid}})/\bar{\epsilon}_{\text{UNet}}$. Reported values are means with corresponding 95% confidence intervals, obtained via error propagation (delta method).

Metric	1:1-Hybrid (%)	1:2-Hybrid (%)	1:3-Hybrid (%)
$\bar{\epsilon}_{\omega}$	75.3 [65.6, 84.9]	82.5 [75.5, 89.2]	81.3 [73.7, 88.9]
$\bar{\epsilon}_{\text{MSE}}$	92.6 [86.3, 98.9]	97.1 [95.0, 99.2]	95.7 [92.1, 99.3]

values should be interpreted as problem-dependent rather than universal performance limits. The UNet surrogate model achieves less runtime reductions than the FNO. This observation is likely related to the higher inference cost of the UNet, illustrating that runtime reduction depends not only on coupling cadence and the relation between ΔT and Δt but also on surrogate evaluation speed rather than its larger total parameter count, indicating that the overall runtime reduction is primarily governed by the prescribed coupling cadence as well as model inference speed. Depending on the hybrid configuration, runtime reductions range from approximately 2.2% to 5.2%.

Taken together, these results demonstrate that hybrid UNet–LBM achieves a favorable balance for accuracy, but lacks in efficiency compared to the FNO–LBM coupling. Despite requiring architectural modifications and substantially more parameters than the FNO, the UNet exhibits the same qualitative response to hybridization, with periodic LBM corrections significantly reducing both short- and long-term prediction errors. The proposed framework therefore appears

Table 18

We compute the relative reduction of simulation runtime $\epsilon_T = (T_{\text{LBM}} - T_{\text{Hybrid}})/T_{\text{LBM}}$ with corresponding 95% confidence intervals for runtime reduction using a paired non-parametric bootstrap with 100,000 resamples and percentile-based bounds. As additional reference, the relative reduction of simulation steps $\epsilon_N = (N_{\text{LBM}} - N_{\text{Hybrid}})/N_{\text{LBM}}$ for each hybrid approach is given compared to pure LBM simulation.

Metric	1:1-Hybrid (%)	1:2-Hybrid (%)	1:3-Hybrid (%)
ϵ_T	5.2 [4.0, 6.3]	3.1 [2.5, 3.6]	2.2 [1.8, 2.6]
ϵ_N	50	34	25

largely surrogate-model agnostic, providing substantial accuracy improvements for both operator-learning (FNO) and conventional convolutional (U-Net) surrogate models.

Data availability

Data generated during the current study is available from the corresponding author upon reasonable request.

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